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AI-generated videos in medical education: systematic review

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ABSTRACT

Background Artificial intelligence (AI)-generated text to video is emerging in medical education, but its effectiveness, accuracy and safety remain uncertain. We aimed to synthesise empirical studies evaluating these tools in learner or patient education.

Methods A comprehensive search was conducted in MEDLINE/PubMed, Google Scholar, Scopus, Cochrane Review and Web of Science for studies published up to January 2025. Eligible studies evaluated AI-generated text to video for medical or patient education, reporting both quantitative and qualitative outcomes. Two reviewers screened and extracted data. The review adhered to Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.

Results Out of 103 identified studies, five studies met the inclusion criteria: four evaluated patient education and one evaluated physician training. Clinical areas were ophthalmology (2/5), plastic surgery (1/5), dysphagia rehabilitation (1/5) and neurosurgical training (1/5). In ophthalmology, control materials outperformed AI-generated videos on image/script accuracy ($p<0.005$), with similar script-image alignment. In dysphagia rehabilitation, a randomised trial reported improvements in swallowing function and related outcomes with an AI-assisted video game intervention ($p<0.001$). A plastic surgery study reported greater user preference for a video avatar tool compared with a text chatbot (63.5% vs 28.1%). Across the reviewed studies, samples were small and CIs were rarely reported. Outcome measures were heterogeneous, and meta-analysis was not feasible.

Conclusion AI-generated videos can enhance engagement or selected outcomes in certain contexts, yet concerns about accuracy and inconsistent measurement persist. Current evidence is sparse and mixed. Currently, these tools can complement, rather than replace, standard resources until non-inferiority is demonstrated for the prespecified primary outcomes.

PROSPERO registration number CRD42025640042.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Artificial intelligence (AI)-generated text-to-video tools are widely promoted as a way to make complex medical information more engaging for learners and patients, but the scientific evidence behind their accuracy, educational benefit and safety has not been systematically synthesised.

WHAT THIS STUDY ADDS

⇒ Our review identified only five empirical studies; while AI-generated videos consistently improved patient or trainee engagement and understanding, all studies also reported content inaccuracies, bias or usability issues, thereby mapping the current evidence gap.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ By cataloguing both benefits and shortcomings, the review provides a baseline for quality benchmarks and highlights the need for rigorous validation and ethical oversight before widespread adoption of AI-generated video tools in medical education and patient communication.

INTRODUCTION

Mastering medicine for providers is long and arduous and requires combining scientific knowledge, emotional intelligence, compassion, curiosity, creativity and clinical expertise.[1 2] This increases the burden on physicians to bridge communication gaps and requires effective ways to teach complex decision-making and procedures.[3–5] Patients who understand their care can make educated decisions, stay engaged and have improved outcomes.[6–8] However, medical terms are often complex, leading to confusion and stress.[9–11]

Generative artificial intelligence (gen AI) is increasingly deployed for various applications in healthcare.[12 13] A new development is AI-generated text to video, such as OpenAI's Sora or Synthesia platform.[14–16] A short overview is provided in online supplemental figure S1. However, despite the widespread use of video in health education, rigorous evaluations of AI-generated text to video, specifically within medical and patient education, remain sparse and methodologically inconsistent. This creates uncertainty about their comparative value and appropriate use.

Nevertheless, prior syntheses largely examine conventional video or generic AI in education and do not determine whether text to video improves learning or accuracy versus standard materials in health contexts, and recent reviews highlight opportunity but also underscore gaps in empirical evidence.[17 18]

The technology can be used to educate health professionals and patients by turning complex medical concepts into clear visual information, as well as understanding current gaps and future implications.[19 20] However, there is minimal current literature compiling the current applications or recognising the implications in medical education. Accordingly, the aim of this study was to synthesise empirical studies of AI-generated text to video for medical and patient education, quantify what is known about learning and accuracy and identify key uncertainties, such as safety, fairness, accessibility and technical reliability, that future research should address to support reproducible evaluation. We also outline actionable measurement parameters and reporting conventions.

METHODS

Information sources and search strategy

We searched in January 2025 to identify studies describing the application of AI-generated text to video for medical education. We searched MEDLINE/PubMed, Google Scholar, Scopus, Cochrane Library and Web of Science for papers published up to January 2025. Full, database-specific strategies with last-run timestamps are provided in the online supplemental material. We also searched grey literature (major preprint servers and relevant conference proceedings) and performed forward and backward citation tracking of included studies and key reviews. Records from all sources were exported for centralised deduplication prior to screening.

Deduplication, screening and inter-rater agreement

Records from all sources were deduplicated deterministically in Rayyan with a manual near-duplicate review (exact DOI matches removed; where DOI was absent, near-identical title+year entries were merged after manual check). Two reviewers independently screened titles/abstracts and full texts against prespecified criteria, with consensus or third reviewer adjudication for disagreements. When outcome data were unclear, we contacted study authors for clarification. If data remained unavailable, the item was recorded as not reported. We did not impute missing values and did not derive unreported statistics.

Eligibility window

We searched from 2010 through January 2025 to ensure transparency, capture any early generative systems and facilitate reproducibility. No eligible studies were identified prior to 2022. All included studies were published between 2022 and 2024. We retained the broader window to avoid time-based selection bias and enable exact replication of our search.

Protocol and registration

Ethical approval was not required, as this is a systematic review of previously published research and does not include individual participant information. Our study followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines. The study was registered with PROSPERO (CRD42025640042) on 18 January 2025 and was last edited on 29 January 2025. The official record is available at https://www.crd.york.ac.uk/prospero/

Inclusion and exclusion process

Publications resulting from the search were initially assessed by two authors (YA and VS) for relevant titles and abstracts. Next, full-text papers underwent an independent evaluation by two authors (EK and BG).

Inclusion criteria

The inclusion criteria were as follows: (1) studies involving medical students, practising healthcare professionals or patients in a medical education context; (2) studies evaluating AI-generated text-to-video technologies used in medical education, such as teaching theoretical concepts, practical skills or patient education; (3) studies comparing AI-generated text-to-video tools with traditional educational methods or other digital education tools; (4) original research such as randomised controlled trials (RCTs), quasiexperimental studies, observational studies, mixed-methods studies and qualitative research; (5) studies published in English; and (6) studies published from 2010 to January 2025, to focus on the recent development of AI-generated video technologies.

Exclusion criteria

The exclusion criteria included: (1) studies focused on non-medical education or general education without specific applications to healthcare or medicine; (2) studies that do not involve AI-generated text-to-video technologies; (3) non-original studies such as opinion pieces, editorials or commentaries without primary data; (4) studies published in languages other than English; and (5) abstract-only publications, dissertations or unpublished theses without access to full data (figure 1). Any study in question was discussed among all authors until reaching a unanimous agreement.

Definitions and outcomes

We defined AI-generated text to video as generative systems that synthesise novel video frames aligned to a script or prompt. Templated slide shows or edited pre-existing footage without generative synthesis were excluded. Primary outcomes included accuracy (both factual and visual) and learning (pre-performance and post-performance, as well as delayed retention). Secondary outcomes included comprehensibility, usability, engagement, accessibility, safety and fairness. Operational definitions and recommended instruments are prespecified in online supplemental table S2 to support reproducible measurement.

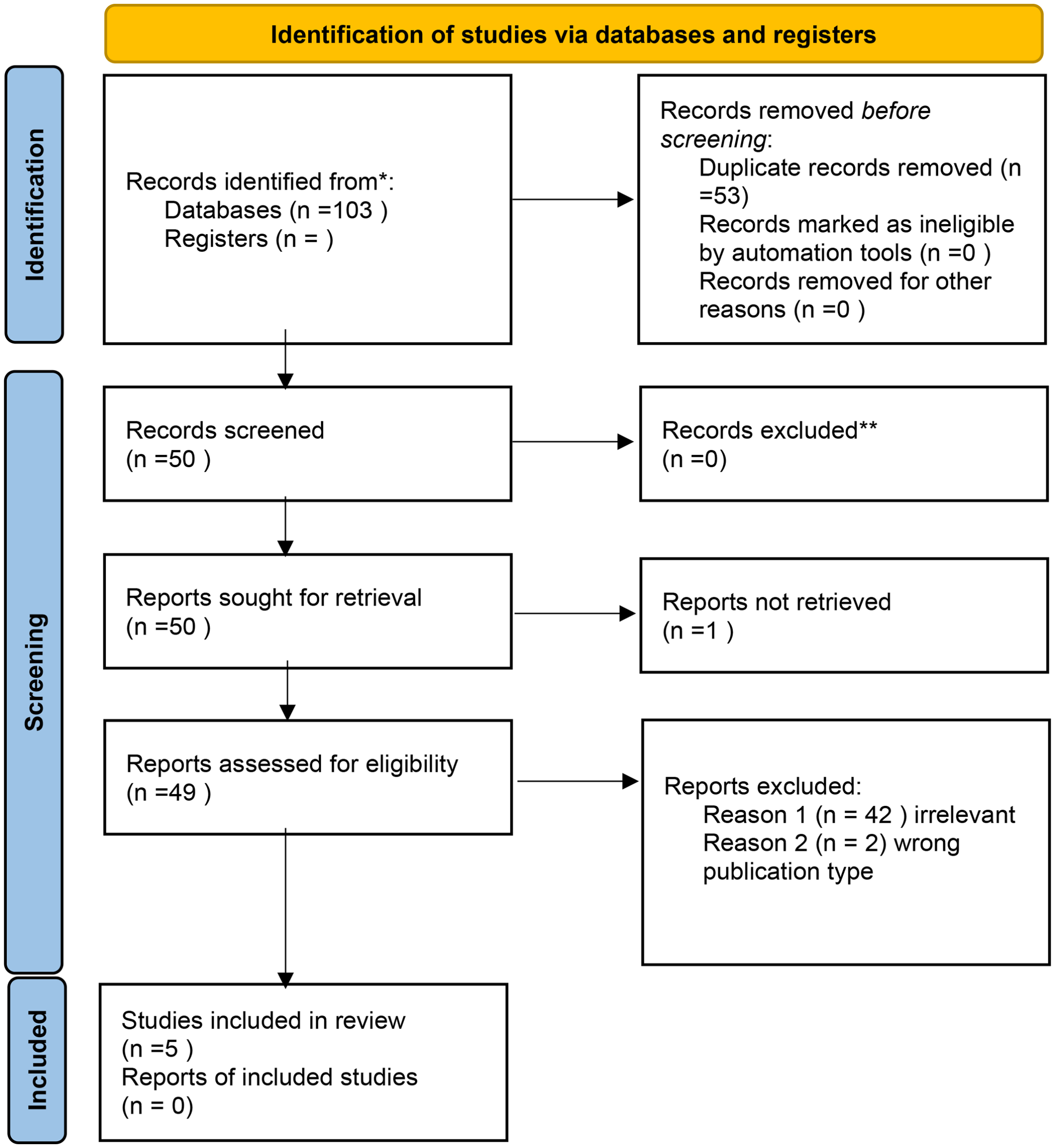

Figure 1 Flow diagram depicting the inclusion and exclusion process of reviewed studies.

Synthesis and meta-analysis plan

If ≥3 clinically and methodologically comparable studies reported the same outcome with harmonised measures, we planned a random-effects meta-analysis using restricted maximum likelihood (REML), applying Hedges' g or mean difference for continuous outcomes and risk ratio (RR) or odds ratio (OR) for binary outcomes, and reporting 95% CIs, τ^2, and I^2. We planned sensitivity analyses excluding high-risk of bias studies and, if ≥10 studies were available, assessment of small-study bias. Because the included studies were few and heterogeneous, no meta-analysis was performed.

Risk of bias assessment

Randomised trials were appraised using the Cochrane Risk of Bias 2 (RoB 2) tool, and non-randomised studies using the Risk Of Bias In Non-randomized Studies – of Interventions (ROBINS-I) tool. Two reviewers assessed the risk of bias independently with consensus resolution. Domain-level judgements and justifications are provided in online supplemental table S1.

Supplementary materials include online supplemental videos 1–14 and Figures S1–S34; full cross-references are listed in the Supplementary Index Table S3.

RESULTS

Study selection and characteristics

All included studies were published between 2022 and 2024. None meeting our criteria were found before 2022. The search yielded 103 studies, out of which 53 were duplicates and 45 were excluded due to publication type and irrelevance to the review subject (figure 1). Out of five relevant studies, four (80%) detailed utilisation of AI-generated text to video for the purpose of patient education, and one (20%) discussed the technology

Table 1 General features of reviewed studies

Study	Journal	Intervention	Medical field
Macri *et al*[21]	*Ophthalmic Research*	Patient education	Ophthalmology
Lawson McLean and Gutiérrez Pineda[22]	*International Journal of Computer Assisted Radiology and Surgery*	Physician education	Neurosurgery
Han *et al*[23]	*Cureus*	Patient education	Ophthalmology
Kim *et al*[24]	*PRS Global Open*	Patient education	Plastic surgery
Zhang *et al*[25]	*Clinical Nutrition*	Patient education	ENT/nutrition

ENT, Ear nose throat.

utilisation for physician medical training. Two studies evaluated applications in ophthalmology, one in neurosurgery, one in plastic surgery and one in rehabilitation (poststroke dysphagia). The studies were conducted in various medical fields and applied different AI tools (table 1).

Descriptive summary of results

Macri *et al*[21] performed a cross-sectional survey (n=15), evaluating patient responses to an AI-generated presenter in educational videos about postoperative face-down positioning after vitreoretinal surgery (no comparator). Outcomes included anxiety (Spielberger State-Trait Anxiety Inventory-6) and perceived quality (Global Quality Score). Effect sizes or 95% CIs were not reported. Descriptive results are presented as percentages in table 2. The main study limitation relevant to interpretation was the absence of a comparator arm and reliance on self-reported measures. Additional limitations are listed in table 3.

Lawson McLean and Gutiérrez Pineda[22] conducted a descriptive feasibility report, exploring the use of OpenAI's video generation model, Sora,[21] to create realistic neurosurgical simulations for trainee education. There was no formal comparator. The study did not report quantitative learning or performance outcomes (eg, skills, knowledge retention), and the results were narrative (table 2). Reported operational constraints (privacy, dataset quality, computation) are itemised in table 3.

Han *et al*[23] performed a non-randomised comparative evaluation of AI-generated ophthalmic surgery videos versus control content. Primary outcomes were accuracy, visual fidelity and related quality metrics. They reported statistical significance (p<0.005) for group differences but did not provide standardised effect sizes or CIs (table 2). Observed technical flaws (eg, fabricated imagery) are detailed in table 3.

Kim *et al*[24] conducted a preference survey (n=396), comparing patient perceptions of a text-based chatbot and a human-like AI VideoBot (Synthesia) for delivering breast reconstruction information. Among 396 women aged 18–64, both assistants provided identical content. Preference for VideoBot was 63.5% vs 28.1%, corresponding to 252 vs 111 respondents (the remaining 33 indicated other/no preference). CIs and p values were not reported (table 2). Responses were self-reported, introducing potential response bias. Limitations are summarised in table 3.

Zhang *et al*[25] conducted an RCT (n=84; 42 per group) in poststroke dysphagia, comparing an AI-based video game (AI-VG) system in a swallowing training system (30 min/day, 5 days/week, 4 weeks) versus usual conventional swallowing exercises. Outcomes included swallowing function (primary), laryngeal function, oral intake, nutritional status and swallowing-related quality of life. Between-group differences favoured AI-VG on swallowing function at postintervention and 1-month follow-up (significant group effects with reported mean differences and 95% CIs; p<0.001), with higher adherence in the intervention group (median 18 (IQR 17–20) vs 16 (IQR 15–17), p<0.001) and higher acceptance scores. Laryngeal function showed no significant difference (table 2). A key limitation was variability in participants' technological proficiency, potentially affecting engagement and treatment adherence (table 3).

Online supplemental figure S2 depicts an overview of the reviewed studies.

DISCUSSION

Platforms like Sora and Synthesia can turn complex medical content into clear and engaging visuals. This innovation holds promise for enhancing both theoretical learning and practical skills. Yet, the technology is new. It needs further validation. Its accuracy, adaptability and real-world use must improve before it can be a reliable educational tool. Notably, learning gains were observed in a randomised setting,[25] while accuracy lapses were documented in ophthalmology evaluations,[23] underscoring that benefits and risks are context dependent rather than universal. Given the small and heterogeneous evidence base (five studies), statements that extend beyond the reported findings are framed as testable hypotheses rather than conclusions. We propose quality parameters and evaluation metrics for AI video application in health professional education in online supplemental table S2. Across the five studies (one RCT and four non-randomised), the evidence is promising but inconsistent: signals of benefit (eg, improved swallowing outcomes in the RCT and higher stated preference) coexist with

Table 2 Summary of study parameters and key results

Study	AI tool	Control tool	Performance metrics	Prompt	General results
Macri *et al*[21]	Synthesia	N/A	Surveys STAI-6 score	N/A	>50% agreed the AI video is clear, helpful and reassuring.
Lawson McLean and Gutiérrez Pineda[22]	Sora	Traditional training methods	Technical skill Error rate Procedure completion time Cognitive load assessment Decision-making quality Feedback from trainees Peer and mentor reviews Preassessment and postassessment Continuous monitoring	'A video simulation of the step-by-step process of resecting a brain tumor, including the approach, craniotomy, and tumor removal.'	Immersive simulations for skill development Exposure to a diverse range of surgical scenarios Standardised, objective methods for assessing and providing feedback on trainee competencies
Han *et al*[23]	InVideo/ClipTalk/EasyVid	Gold standard control videos from the AAO	3-point scale of four-category quality assessment: image accuracy, script accuracy, image clarity, script alignment with the video	'Create a patient-centered educational video depicting a step-by-step guide on what to expect during LASIK, PRK, or SMILE surgery. Be as anatomically and surgically accurate as possible.'	Medical accuracy: Control>InVideo, EasyVid and ClipTalk for LASIK, PRK and SMILE ($p<0.005$). Video quality and resolution: no significant differences were found ($p>0.05$). Script accuracy: Control>ClipTalk for the SMILE instructional video ($p<0.05$). Script alignment: AI-generated videos had comparable alignment between the script and the images being displayed compared with the control videos. Total score: control videos outperform all three AI text-to-video generators.
Kim *et al*[24]	Synthesia	Automated text-based chatbot integrated with ChatGPT 3.5	1–7 Likert scale rating 16-item survey	N/A	No difference in perceptions between the two platforms. Participants preferred the VideoBot over the traditional chatbot (63.5% vs 28.1%).

Continued

Table 2 Continued

Study	AI tool	Control tool	Performance metrics	Prompt	General results
Zhang *et al*[25]	AI-VG	Usual care treatment: lip, tongue and chin exercises	Swallowing function measured by GUSS SSA FOIS MNA-SF SWAL-QOL TAM	N/A	AI-VG group had significantly improved GUSS scores (p<0.001). AI-VG group had lower SSA scores with no statistical difference (p=0.100). AI-VG group showed an additional increase in the FOIS scores (p<0.001). AI-VG group had a significantly improved MNA-SF score (p=0.034). AI-VG group showed a more significant decrease in SWAL-QOL (p<0.001). Only the AI-VG group completed the acceptance and satisfaction questionnaires. Score 73.00 (72.00–74.00).

AAO, American Academy of Ophthalmology; AI, artificial intelligence; AI-VG, artificial intelligence-based video game; FOIS, Functional Oral Intake Scale; GUSS, Gugging Swallowing Screen; LASIK, laser-assisted in situ keratomileusis; MNA-SF, Mini-Nutritional Assessment Short Form; PRK, photorefractive keratectomy; SMILE, small incision lenticule extraction; SSA, Standard Swallowing Assessment; STAI-6, Spielberger State-Trait Anxiety Inventory-6; SWAL-QOL, Swallowing Quality-of-Life Questionnaire; TAM, technology acceptance model.

limitations (eg, inferior visual accuracy vs controls, sparse effect sizes/CIs and minimal assessment of safety and fairness).

These quality parameters can be actionable with measurable endpoints and clear reporting. For instance, accuracy and visual script alignment, which are primary desired outcomes, can be set with predefined thresholds. Learning can be evaluated using pretesting and post-testing, as well as delayed retention. In addition, cognitive load can be assessed with validated scales, with

Table 3 Limitations presented in reviewed studies

Study	Technical limitation	Evaluation limitation	Performance limitation
Macri *et a*[21]	N/A	No comparison with traditional methods of information delivery.	N/A
Lawson McLean and Gutiérrez Pineda[22]	Implementing transformer architecture requires significant processing power and specialised hardware.	No standardised metrics to evaluate the quality and educational value of AI-generated simulations.	Generated scenarios may lack the unpredictability and complexity of actual surgical situations.
Han *et al*[23]	N/A	No comparison between AI-generated videos and traditional educational methods.	Non-surgical ophthalmic video clips Random clips from different surgical specialties Completely factitious details
Kim *et al*[24]	No flexibility in asking the VideoBot other questions due to incomplete integration of ChatGPT into the Synthesia.ai platform.	N/A	Difficulty for some patients to process information in a video setting as opposed to reading at their own pace.
Zhang *et al*[25]	N/A	Different educational backgrounds can affect comfort and competency with technological interfaces.	N/A

AI, artificial intelligence.

monitoring of safety through error rates and content provenance disclosure and oversight. To support reproducibility, we specify how each metric can be operationalised in online supplemental table S1, and recommend predefined performance margins with absolute difference triggers that prompt mitigation when thresholds are exceeded. Tracking costs and the time needed to update content when guidance changes can assist in cost-effective deployment.

Time is a precious resource. Creating medical education content is resource and time intensive. Gen AI can generate high-quality content rapidly. Patient views and decisions are often influenced by sex, gender, race and ethnicity.[26 27] AI can personalise and adapt videos to different sociodemographic groups.[28] This may help patients feel more at ease with the content or identify with the AI avatar presenting the information. Preference signals for avatar-based delivery[24] suggest potential engagement advantages, while patient-reported measures[21] indicate acceptability in patient-facing contexts; however, both designs were non-comparative or self-reported, so future work should prospectively stratify outcomes by sex, gender, age, language and other relevant factors, and test for parity. For example, we used the prompt mentioned in Han *et al*[23] on Synthesia as an illustrative case. Users choose the avatar's appearance (online supplemental figure S3), voice and background (online supplemental figure S4). To evaluate whether such personalisation benefits learning equitably, studies should predefine subgroup analyses with non-inferiority or minimal important difference margins and report any absolute gaps that would trigger mitigation.

Further, the ability to deliver visual information can increase engagement and improve retention,[19 20] and also serve as a mediator for patients. This is especially true for complex subjects like surgical procedures and diagnostic reasoning. Another important feature of AI video is accessibility, which can potentially allow global access to standardised, up-to-date medical knowledge, bridging educational gaps in resource-limited settings.[29] Evidence for these benefits in our corpus is preliminary and indirect. Accordingly, we recommend reporting accessibility using established criteria (eg, caption quality, language coverage, device compatibility) alongside engagement indicators such as completion and interaction rates, so that pedagogical benefit is interpreted in light of access quality.

AI-generated videos face challenges in accuracy, reliability and trust.[30] Evidence from the included studies shows that accuracy lapses are concrete rather than hypothetical. As demonstrated in Han *et al*,[23] fabricated ophthalmic imagery (eg, laser beams rendered from surgical lamps) highlights risks in visual fidelity even when scripts are broadly correct, and our replication of the same prompts produced spelling errors (online supplemental figures S5 and S6). In contrast, Zhang *et al*[25] reported learning improvements in a randomised setting, indicating that benefits are possible when content is validated and embedded in structured programmes. To reduce error propagation, we recommend prespecified accuracy and safety endpoints, expert auditing reported as errors per video minute with inter-rater reliability and time-stamped versioning so guideline-driven corrections are transparent and auditable. These mixed findings caution against broad generalisations at this stage. Bias is another concern.[31] If these models rely on Western-centric medical literature or datasets that lack diversity, medical education videos could perpetuate biased medical knowledge. This may result in over-representation of certain conditions and treatments relevant to high-income populations, while neglecting diseases that disproportionately affect lower income or minority communities, creating exclusionary medical education. In our corpus, fairness and representation were seldom measured (eg, preference without parity analyses in Kim *et al*[24]; patient-reported perceptions without a comparator in Macri *et al*[21]), which we identify as a gap (tables 2 and 3). Accordingly, we suggest disclosing dataset documentation, reporting avatar representation and presenting outcome parity analyses across key demographic groups. When gaps are detected, remediation such as prompt redesign, asset substitution or expert renarration should be documented. Given the absence of parity analyses in the included studies, equity statements here are prospective hypotheses to be tested. Evaluations should also report outcome parity by sex, gender, age and language, and stratify by device class (including low/mid-range hardware) so equity and access are visible at evaluation time. Ethical considerations must also be addressed, as illustrated by the implementation constraints in Lawson McLean and Gutiérrez Pineda[22] and the fabricated imagery observed in Han *et al*.[23] AI-generated videos could be misused to fabricate content, raising concerns about deception and authenticity in medical education.[32] Moreover, AI can potentially stray from human ethical guidelines and values.[33] Also, patients' use of AI requires cybersecurity measures to ensure that sensitive medical information is secure from unauthorised access, and to protect against adversarial attacks.[34–37] Misuse of AI-generated videos or misleading content will necessitate rigorous debates on accountability and become major areas of regulatory focus. Reporting of consent, privacy and provenance safeguards, including deidentification of any patient-like content, is a possible mitigation, as well as applying tags for synthetic content with documented review by institutional or professional oversight where applicable.

Another concern is the unequal distribution of resources for AI implementation, especially in low-resource settings. Deploying AI text-to-video systems can be expensive, making them inaccessible to institutions with limited budgets.[38] In addition, current AI text to video may not support all languages, limiting its utility in multicultural and global settings.[39]

The high costs and unequal distribution of resources widen the digital divide. Wealthier institutions may benefit from AI-enhanced education, while resource-limited ones struggle with lack of access. Although our included studies did not directly evaluate these outcomes, feasibility observations (Lawson McLean and Gutiérrez Pineda[22]) highlight operational constraints. We therefore identify equity and access as implementation questions to be tested prospectively rather than as established effects. To promote equitable adoption and in line with gaps observed across our five studies, we encourage reporting results stratified by device class (low/mid-range hardware) and by language, together with failure modes and resource requirements, so cost and access considerations are visible at evaluation time.

Medical educators should prepare for these potential adverse implications alongside the benefits and ensure that formal medical education will not suffer in quality and credibility. Depiction of strength versus limitations is presented in online supplemental figure S7. Institutions should define minimum acceptable thresholds before deployment and include plans for periodic re-evaluation and rapid update when clinical guidance changes. In this review, we analysed the literature on AI-generated text to video in medical education. This technology can be used to educate both patients and physicians. Interactive learning platforms, virtual reality and chatbots have already advanced medical education.[40–43] Text-to-video systems may be the next step contingent on stronger evidence from comparative trials. They can improve engagement, accessibility and the visualisation of medical concepts. However, these tools also bring challenges. This includes inaccurate information.[44] Overly relying on them can reduce opportunities for critical thinking, mentorship and interpersonal learning.[45] Finally, one key challenge in using AI as a liaison between doctors and patients is its inability to fully replace human interaction and compassion, particularly in emotionally charged situations.[46 47] Accordingly, and consistent with our findings across the five studies, AI-generated videos should complement rather than replace traditional methods until non-inferiority is demonstrated on prespecified primary outcomes and value is shown on secondary metrics. While large language models can produce empathic language,[48] none of the included studies measured empathy or interpersonal outcomes; therefore, we cannot conclude about empathy from this review. Current AI avatars may not consistently convey non-verbal cues (eg, facial expressions), which remains an open question for future trials.[49 50] These limitations can make AI-generated videos inappropriate for sensitive topics such as prognosis or end-of-life care. In our replication of Han *et al*'s[23] prompts across four platforms, we observed spelling and rendering errors, reinforcing the need for prospective accuracy audits before clinical deployment. (Online supplemental videos 1–11 depict a synthetic, AI-generated avatar; no real person is shown.) The creation process and resulting videos and images further demonstrate these limitations in online supplemental table S2 and figures S8 and S9.

Mitigation strategies

To address these risks and challenges, several steps can take place. Accuracy checks and expert review protocols can be established. To minimise bias, diverse and representative datasets should be used. Data security can be addressed with encryption and adherence to privacy regulations. Future models can support more languages and work in low-resource settings. This is further explored in online supplemental table S1, detailing quality parameters and evaluation metrics we propose for AI video application in medical education (online supplemental table S2).

By addressing these issues, AI text-to-video technology can be used more safely and effectively in medical education.

Key considerations for medical educators

Medical educators should strive to develop their knowledge and skills in this technology, ensuring AI-generated content aligns with evidence-based practices and maintains educational quality. As of today, AI text-to-video tools should complement, not replace, traditional teaching methods. Creating a hybrid model that balances both technology and human expertise would best benefit educators as well as patients. Educators must also address potential biases, ethical concerns and disparities in access to AI tools, particularly in resource-limited settings. By maintaining high standards and critically evaluating AI-generated content, we can responsibly integrate these tools to enhance learning while preserving essential human elements such as mentorship and critical thinking.

Limitations

This review has limitations. AI text-to-video application in medicine and medical education is a new technology; hence, our search did not yield many relevant studies. Due to heterogeneity in study design and data, a meta-analysis was not performed. Future research will help us better understand its integration into medical education. Since the search was conducted until January 2025, new research might have been published and is not included in our review.

CONCLUSION

AI-generated text to video is an emerging tool in medical education, offering scalability, personalisation and engagement. However, accuracy, bias and ethical concerns must be addressed. Future research should prioritise validation through comparative studies. Educators should use AI video to complement, not replace, traditional teaching. While AI evolves quickly, it cannot yet replace human educators. In the words of Plutarch, "the mind is not a vessel to be filled but a fire to be kindled," and that is still humans' greatest advantage. Based on the five included studies, the current evidence is promising but inconsistent. Until non-inferiority to standard materials is demonstrated on prespecified primary outcomes, and accuracy, safety and fairness are transparently reported, AI-generated videos should remain adjuncts rather than replacements.

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Contributors YA conducted the literature search and data synthesis and wrote the primary manuscript. VS, BG, PK and DT reviewed and edited the manuscript. GNN and EK contributed to the conceptualisation, data synthesis and editing of the manuscript. All authors read and approved the final manuscript. EK is the guarantor of this work. We confirm that artificial intelligence tools were used in the following way. In study materials, we generated illustrative, non-identifying exemplar content using text-to-video platforms to demonstrate prompt-to-video workflows referenced in the manuscript and supplement (avatar clips and screenshots). These exemplars did not involve real patients or clinicians, and no patient data were used. All AI-assisted content was reviewed, verified and edited by the authors, who take full responsibility for the content and for verifying the accuracy of references, data and interpretations. No confidential, proprietary or patient-level information was input into any AI system. AI tools were not used for data extraction, risk-of-bias judgements, statistical analysis or concluding.

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